

UNIT-IV

LEETCODE EXERCISE

1.Delete Duplicate Emails

Table: Person

Column Name	Type
id	int
email	varchar

Id is the primary key (column with unique values) for this table.

Each row of this table contains an email.

The emails will not contain uppercase letters.

Q:Write a solution to delete all duplicate emails, keeping only one unique email with the smallest id.

Solution

 Code

MySQL   Auto

```
1 delete p1
2 from Person p1, Person p2
3 where p1.Email = p2.Email and p1.Id > p2.Id
```

✓ Testcase | > Test Result

Accepted Runtime: 72 ms

✓ Case 1

Input

Person =

```
| id | email |
| -- | - - - - - |
| 1 | john@example.com |
| 2 | bob@example.com |
| 3 | john@example.com |
```

Output

```
| id | email |
| -- | - - - - - |
| 1 | john@example.com |
| 2 | bob@example.com |
```

Explanation:

john@example.com is repeated two times. We keep the row with the smallest Id = 1 .

2. Customers Who Never Order

Suppose that a website contains two tables,
the Customers table and the Orders table.

Write a SQL query to find all customers who never order anything .

Table: Customers.

Id	Name
1	Joe
2	Henry
3	Sam
4	Max

Table: Orders.

Id	Customer Id
1	3
2	1

Using the above tables as example, return the following:

Customers
Henry
Max

Solution

 Code

MySQL   Auto

```
1 select c.Name as Customers
2 from Customers c
3 where c.Id not in (select CustomerId from Orders);
```

☒ Testcase  Test Result

Accepted Runtime: 111 ms

 Case 1

Input

Customers =

id	name
1	Joe
2	Henry
3	Sam
4	Max

Orders =

id	customerId
1	3
2	1

Explanation

- SELECT c.Name → Customer names are selected.
- FROM Customers c → Data is taken from the Customers table.
- SELECT CustomerId FROM Orders → Finds customers who have placed orders.
- NOT IN → Finds customers who did not place any order.

3.Biggest Single Number

Table my_numbers contains many numbers in column num including duplicated ones. Can you write a SQL query to find the biggest number, which only appears once.

Num
8
8
3
3
1
4
5
6

For the sample data above, your query should return the following result:

num
6

Solution

</> Code

MySQL   Auto

```
1 select max(num) as num
2 from
3     (select num
4      from my_numbers
5      group by num
6      having count(num) = 1) n
```

Output

```
| num |  
| --- |  
| 6   |
```

Explanation:

- The single numbers are 1, 4, 5, and 6.
Since 6 is the largest single number, we return it.

4.Game Play Analysis I

Table: Activity

Column Name	Type
player_id	int
Device_id	int
Event_date	date
Games_played	int

(player_id, event_date) is the primary key of this table.

This table shows the activity of players of some game.

Each row is a record of a player who logged in and played a number of games (possibly 0) before logging out on some day using some device.

Q:Write an SQL query that reports the first login date for each player.

The query result format is in the following example:

Solution

</> Code

MySQL   Auto

```
1 select a.player_id, min(a.event_date) as first_login
2 from Activity a
3 group by a.player_id
```

☒ Testcase |  Test Result

Accepted Runtime: 69 ms

☒ Case 1

Input

Activity =

player_id	device_id	event_date	games_played
1	2	2016-03-01	5
1	2	2016-05-02	6
2	3	2017-06-25	1
3	1	2016-03-02	0
3	4	2018-07-03	5

Output

player_id	first_login
1	2016-03-01
2	2017-06-25
3	2016-03-02

Explanation

- GROUP BY player_id: This collects all rows belonging to the same player into a single group.
- MIN(event_date): Since dates are ordered chronologically, MIN() finds the earliest date in each player's group, which represents their first ever login.
- AS first_login: Renames the calculated column to match the required output

5.Exchange Seats

Table: Seat

Column name	Type
id	int
student	varchar

id is the primary key (unique value) column for this table.

Each row of this table indicates the name and the ID of a student.

The ID sequence always starts from 1 and increments continuously.

Q:Write a solution to swap the seat id of every two consecutive students. If the number of students is odd, the id of the last student is not swapped.

Return the result table ordered by id in ascending order.

The result format is in the following example.

 Code

MySQL   Auto

```
1 SELECT
2     a.id id,
3     COALESCE(b.student, a.student) student
4 FROM
5     seat a LEFT JOIN seat b
6     ON b.id = (CASE
7         WHEN a.id%2=1 THEN a.id+1
8         ELSE a.id-1
9         END)
10 ORDER BY a.id
```

✓ Testcase | ➤ Test Result

Accepted Runtime: 73 ms

✓ Case 1

Input

Seat =

id	student
1	Abbot
2	Doris
3	Emerson
4	Green
5	Jeames

Output

id	student
1	Doris
2	Abbot
3	Green
4	Emerson
5	Jeames

Explanation:

- Note that if the number of students is odd, there is no need to change the last one's seat.

6. Triangle Judgement

A pupil Tim gets homework to identify whether three line segments could possibly form a triangle.

However, this assignment is very heavy because there are hundreds of records to calculate.

Could you help Tim by writing a query to judge whether these three sides can form a triangle, assuming table triangle holds the length of the three sides x, y and z.

X	Y	Z
13	15	13
20	20	15

For the sample data above, your query should return the follow result:

X	Y	Z	Triangle
13	15	13	No
20	20	15	Yes

Solution

 Code

MySQL   Auto

```
1 select *,
2     if(x+y>z and x+z>y and y+z>x, 'Yes', 'No') as
3     triangle
4 from triangle
```

☒ Testcase  Test Result

Accepted Runtime: 83 ms

☒ Case 1

Input

Triangle =

x	y	z
13	15	13
20	20	15

Output

x	y	z	triangle
13	15	13	No
20	20	15	Yes

Explanation

- GROUP BY player_id groups each player's records together, and MIN(event_date) picks the earliest date from each group as their first login

7. Consecutive Numbers

Table: Logs

Column name	Type
id	int
num	varchar

In SQL, id is the primary key for this table.

id is an autoincrement column starting from 1.

Q:Find all numbers that appear at least three times consecutively.

Return the result table in any order.

The result format is in the following example.

 Code

MySQL   Auto

```
1 SELECT DISTINCT L1.NUM ConsecutiveNums
2   FROM Logs l1, Logs l2, LOGS L3
3  WHERE l1.Id = l2.Id - 1
4        AND l2.ID=L3.ID - 1
5        AND l1.Num = l2.Num
6        AND L2.NUM = L3.NUM;
```

☒ Testcase | [> Test Result](#)

Accepted Runtime: 90 ms

☒ Case 1

Input

Logs =

id	num
1	1
2	1
3	1
4	2
5	1
6	2
7	2

Output

ConsecutiveNums
1

Explanation:

- 1 is the only number that appears Consecutively for at least three times.